# Cut-sets and Cut-Vertices

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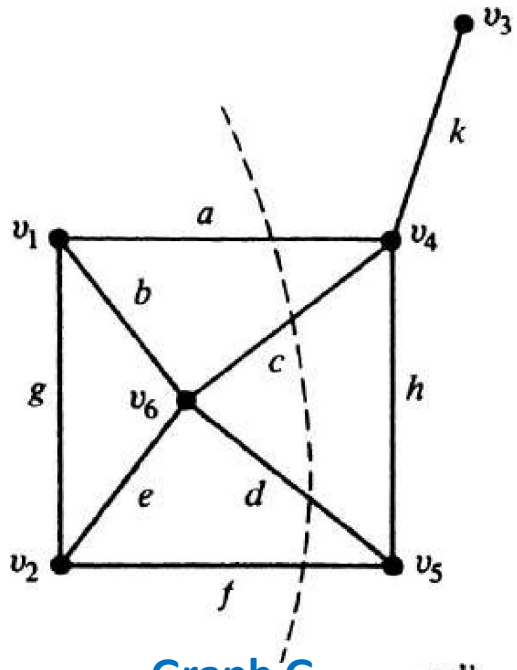
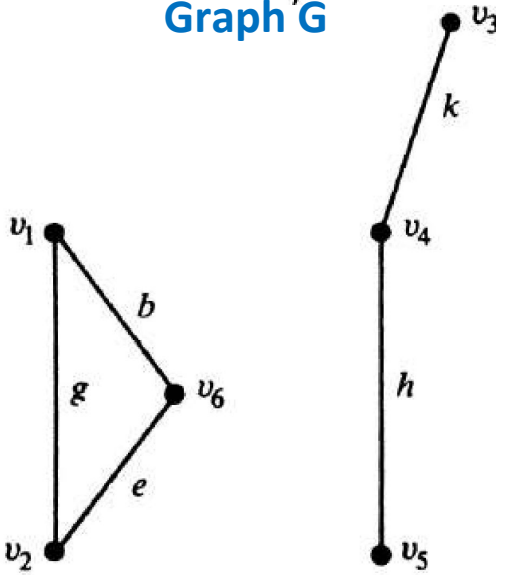
# Cut-set of a Graph

- A **cut-set** is a **set of edges** of a **connected graph G**, **whose removal leaves G disconnected**, provided removal of **no proper subset** of these edges disconnects G.
  - For e.g., in graph **G** the **set of edges** {a, c, d, f} is a **cut-set**. There are **many other cut-sets**, such as {a, b, g} , {a, b, e, f} , {d, h, f}, {k} etc. The **set of edges** {a, c, h, d} , is **not a cut-set**, **because** one of **its proper subsets**, {a, c, h}, is **a cut-set**.
- As **no proper subset** of a **cut-set** can be a **cut-set**, **a cut-set** is also called a **minimal cut-set**, a **proper cut-set**, or a **simple cut-set**.
- A **cut-set** is a **minimal number of edges** whose **removal from G destroys all paths** between **two sets** of **vertices**.
- A **cut-set** always **cuts** a **graph** into **two** or equivalently **increase component** by **one**. Therefore, a **cut-set** is also **defined** as a **minimal set** of **edges** in a **connected graph** whose **removal reduces** the **rank** (n - k) of the graph by **one**.




Graph G



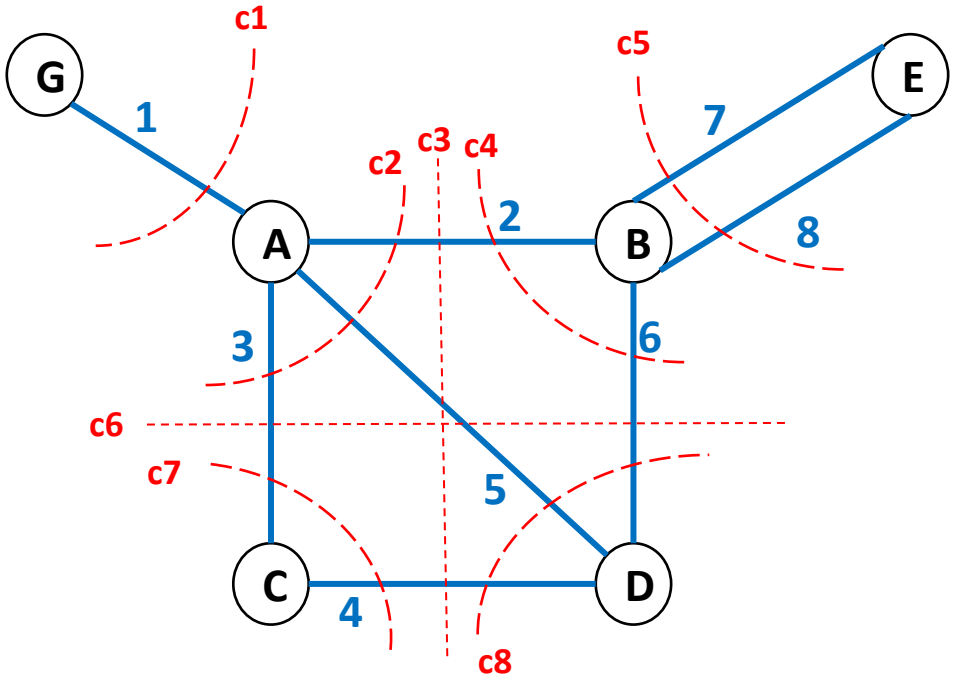

**Removal of a cut-set {*a*, *c*, *d*, *f*} from a graph G cuts it into two**

# Cut-set Matrix

- **cut-set matrix**: each **row** **represents** a **cut-set**, **C(i,j)=1 represents $j^{th}$ edge** is an **element** of **$i^{th}$ cut-set**.
  - c1={1}, c2={2 ,3, 5}, c3={2, 4, 5}, c4={2, 6}, c5={7, 8}, c6={3, 5, 6}, c7={3, 4}, c8={4, 5, 6}

| C | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| c1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| c2 | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 0 |
| c3 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 0 |
| c4 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 |
| c5 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| c6 | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 0 |
| c7 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 |
| c8 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 |

cut-set matrix



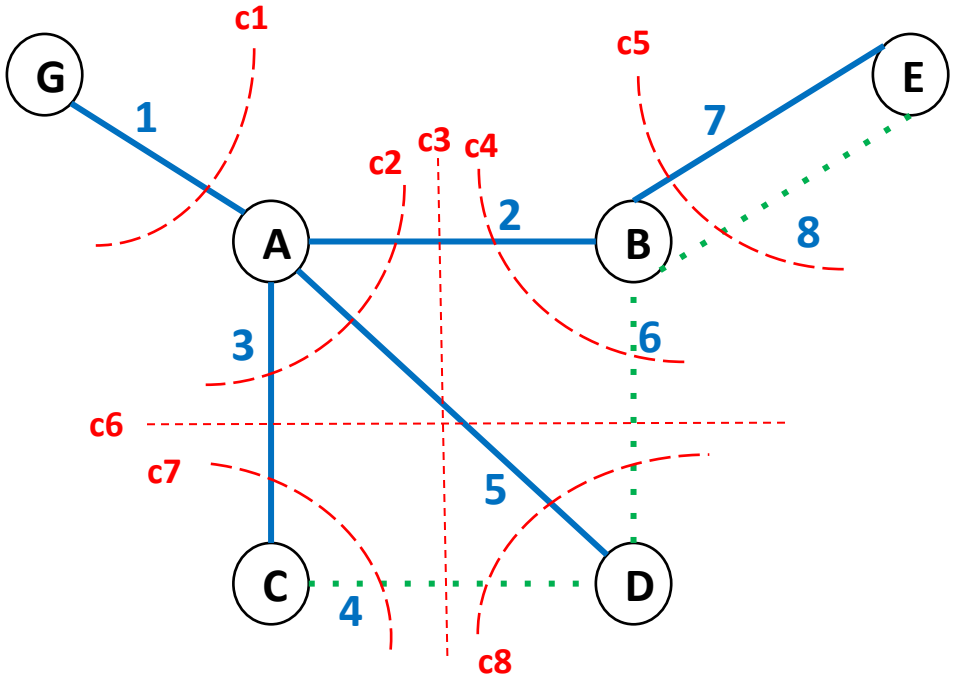

A graph and its cut-sets

# Fundamental Cut-set Matrix

- **Fundamental** (also called **basic**) **cut-set** with respect to a **spanning tree** is a cut-set that **contains exactly one edge** of the **spanning tree**.
  - For the spanning tree {1,2,3,5,7}, c1={1}, ~~c2={2 ,3, 5}~~, ~~c3={2, 4, 5}~~, c4={2, 6}, c5={7, 8}, ~~c6={2, 5, 6}~~, c7={3, 4}, c8={4, 5, 6}

| | 4 | 6 | 8 | 1 | 2 | 3 | 5 | 7 |
|---|---|---|---|---|---|---|---|---|
| c1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| ~~c2~~ | ~~0~~ | ~~0~~ | ~~0~~ | ~~0~~ | ~~1~~ | ~~1~~ | ~~1~~ | ~~0~~ |
| ~~c3~~ | ~~1~~ | ~~0~~ | ~~0~~ | ~~0~~ | ~~1~~ | ~~0~~ | ~~1~~ | ~~0~~ |
| c4 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 |
| c5 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| ~~c6~~ | ~~0~~ | ~~1~~ | ~~0~~ | ~~0~~ | ~~0~~ | ~~1~~ | ~~1~~ | ~~0~~ |
| c7 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| c8 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 0 |

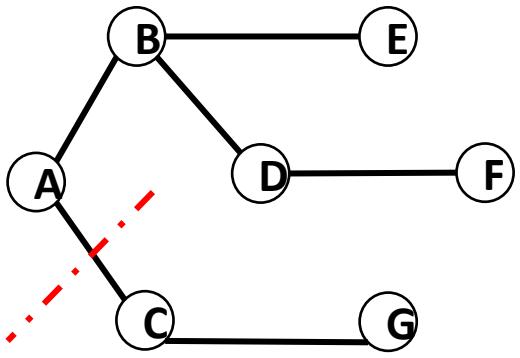
cut-set matrix




A graph and its cut-sets

# Cut-set of a Tree

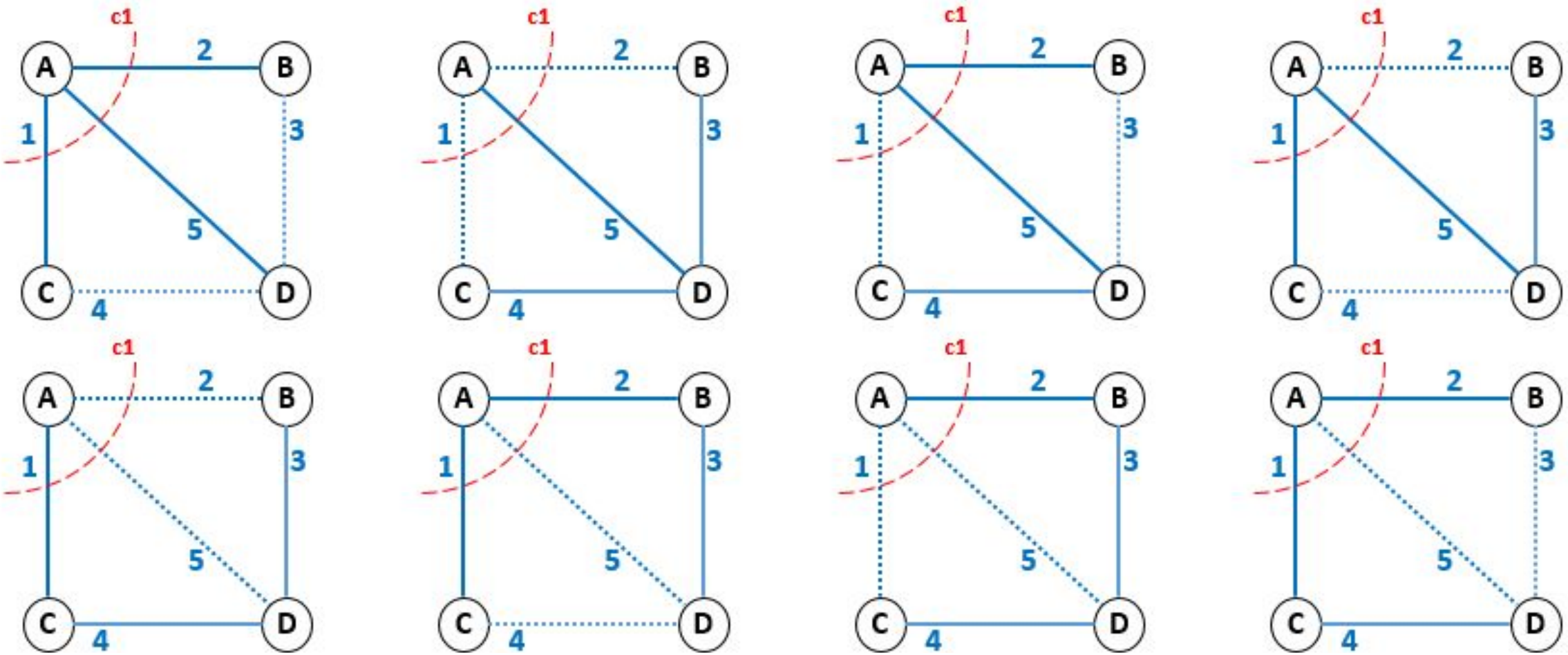
- Since **removal** of **any edge** from a **tree** **breaks** the **tree into two parts**, **every edge** **of a tree** is a **cut-set**.




**Tree in which every edge is an cut-set**

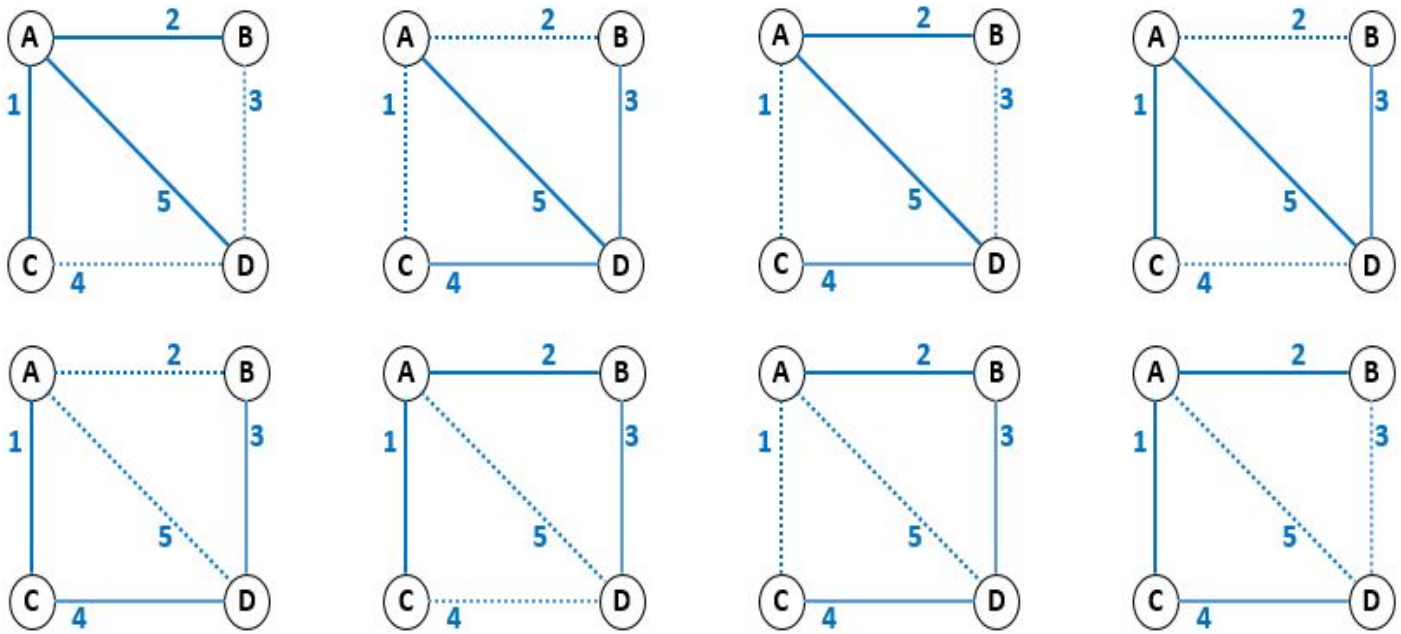
# Properties of Cut-set

- **Every cut-set** in a **connected graph G must contain at least one branch** of **every spanning tree** of **G**.



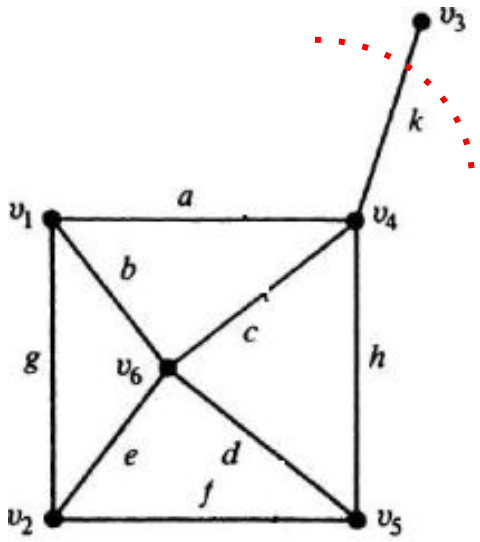
# Properties of Cut-set

- In a **connected graph G**, any **minimal set of edges containing at least one branch** of **every spanning tree** of **G** is a **cut-set**.
  - Let us have the set of edges Q={1,2,5}, a minimal set of edges containing at least one branch of every spanning tree of G.
  - The subgraph $G - Q$ that remains after removing the edges in $Q$ from $G$ is disconnected. Since the subgraph $G - Q$ contains no spanning tree of G.
  - Also, since $Q$ is a minimal set of edges with this property, any edge $e$ from $Q$ returned to $G - Q$ will create at least one spanning tree.
  - Thus the subgraph $G - Q + e$ will be a connected graph.
  - Therefore, $Q$ is a minimal set of edges whose removal from $G$ disconnects G. This, by definition, is a cut-set.

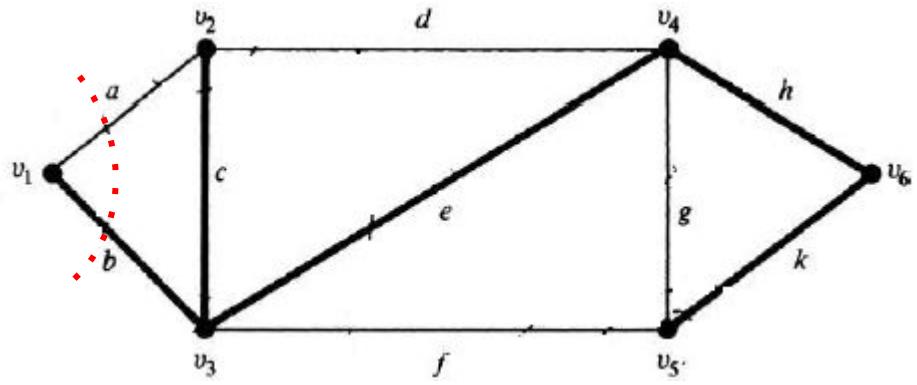


# Edge Connectivity of a Graph

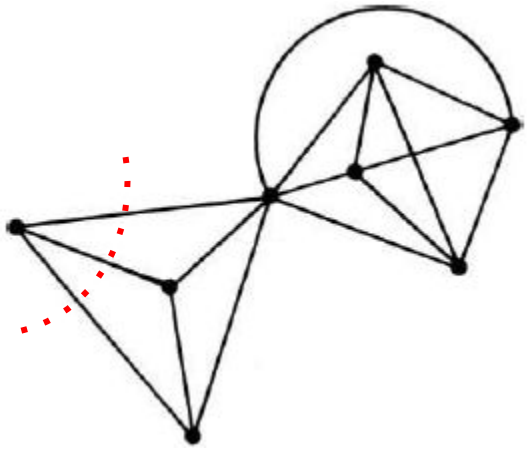
- **Edge Connectivity**: is defined **as** the **number of edges** in the **smallest cut-set** of the **graph G** OR the **minimum number of edges** whose removal (i.e., deletion) **reduces** the **rank of the graph** by **one**.
- The **edge connectivity** of **any tree**, is **one**. The edge connectivities of the graphs **G1**, **G2** and **G3** are **one**, **two**, and **three**, respectively.




Graph G1



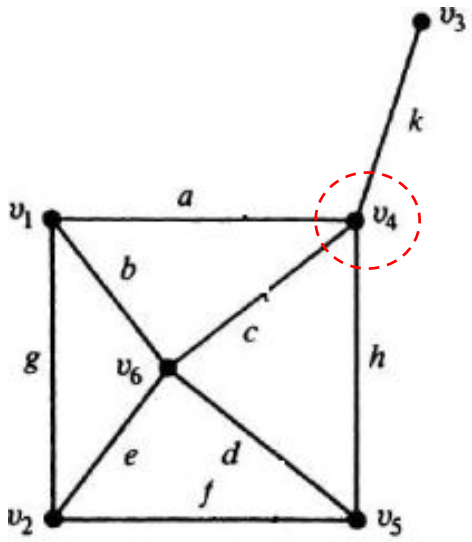

Graph G2



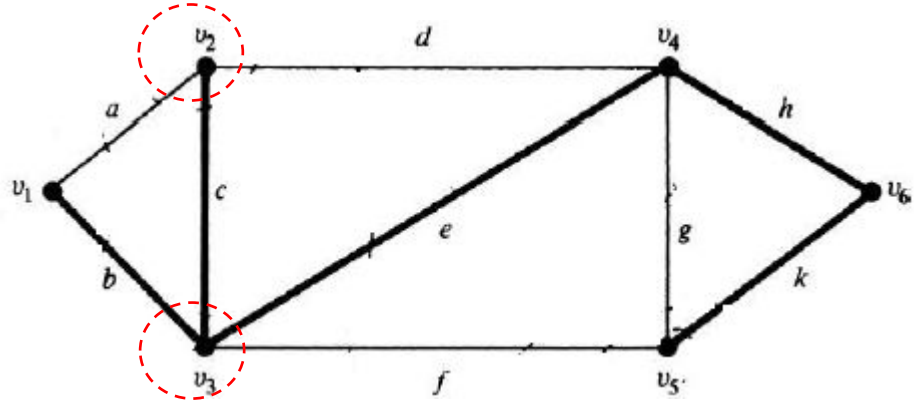
Graph G3

# Vertex Connectivity of a Graph

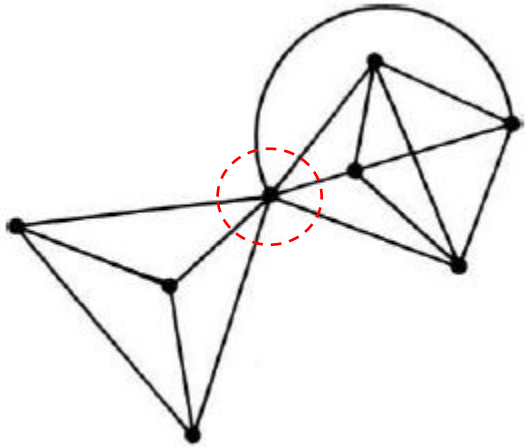
- **Vertex connectivity** (or simply connectivity) of a connected graph **G** is defined as the **minimum number of vertices** whose **removal** from **G** **leaves the remaining graph disconnected**. Note that the vertex connectivity is meaningful only for graphs that have three or more vertices and are not complete.
- The **vertex connectivity** of **any tree**, is **one**. The **vertex connectivity** of the graphs **G1**, **G2** and **G3** are **one**, **two**, and **one**, respectively. The **vertex connectivity** of a **tree** is **one**.




Graph G1



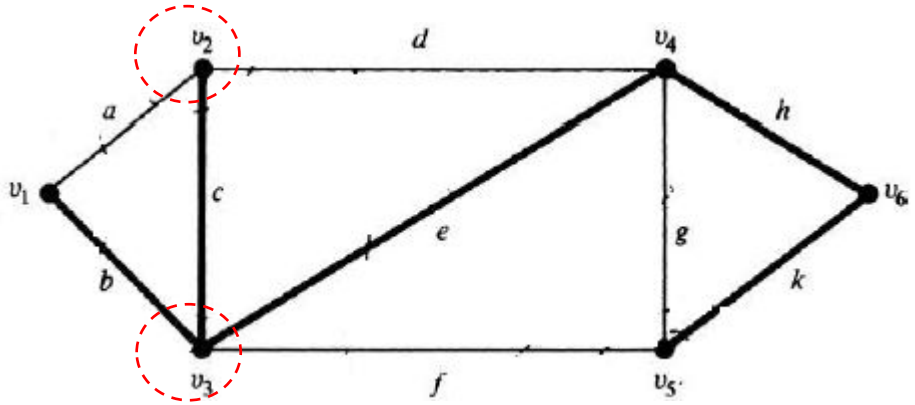

Graph G2



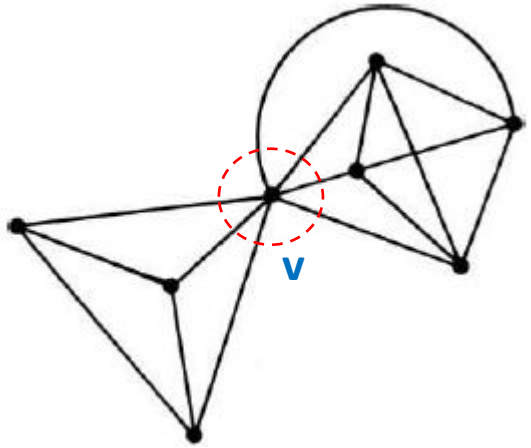
Graph G3

# Separable Graph and Cut-Vertex

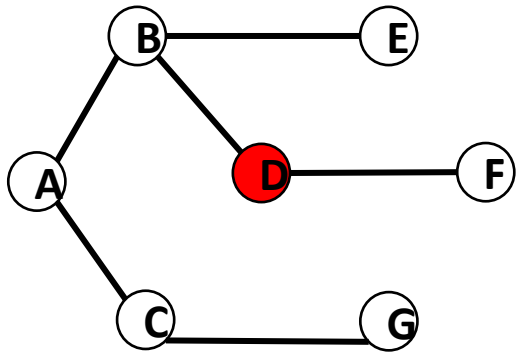
- **Separable Graph**: A **connected graph** is said to be **separable** **if its vertex connectivity** is **one**. **All other** connected graphs are called **non-separable**. Graph **G2** is **separable** whereas **G1** is **non-separable**.
- In a **separable graph** a **vertex** **whose removal disconnects** the **graph** is **called** a **cut-vertex**, a **cut-node**, **or** an **articulation point**. For example, in Graph **G2** the **vertex v** is a **cut-vertex**.
- In a **tree** **every vertex** with **degree** **greater than one** is a **cut-vertex**.




Graph G1



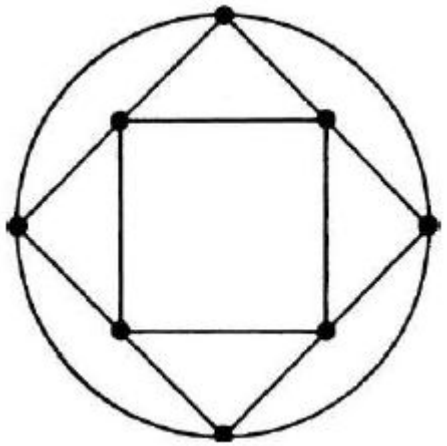

Graph G2



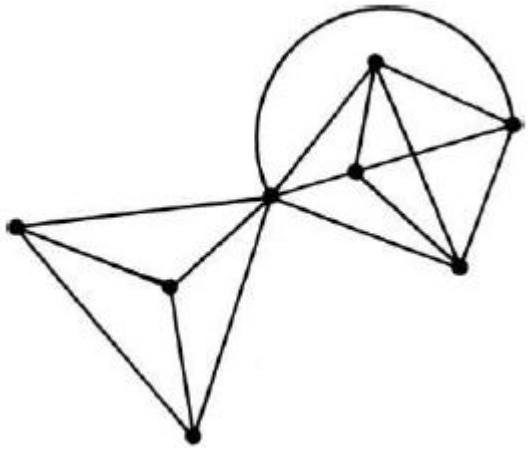

Tree T

# Application of Edge and Vertex Connectivity

- Given **n stations to be connected** by means of **e lines** (**telephone lines**, **bridges**, **railroads**, **tunnels**, or **highways**) where $e \geq n - 1$. What is the best way of connecting? The network should be as invulnerable to destruction of individual stations and individual lines as possible. In other words, **construct a graph** with **n vertices** and **e edges** that **has** the **maximum possible edge connectivity** and **vertex connectivity**.
- **Graph G1** is **better** than **G2** in terms of edge connectivity and vertex connectivity, **even though both** are **with same number of edges** and **vertices**.



**Graph G1** Graph with 8 vertices and 16 edges



**Graph G2** Graph with 8 vertices and 16 edges